

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

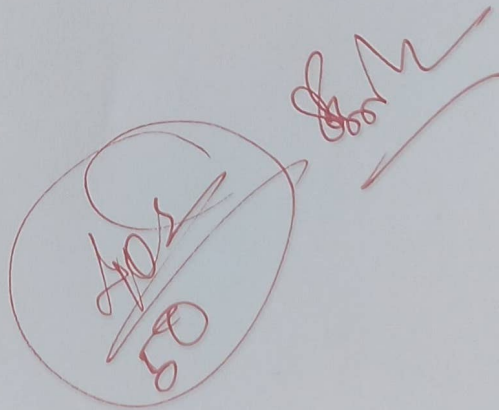
CONCEPT MAPPING

Code of Conduct:

I K. Lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Lahari



CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them. Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Classical Encryption Schemes

Classical Encryption Schemes

Substitution Techniques
(Replace characters)

- * Caesar cipher
- * Monoalphabetic cipher
- * Vigenere cipher
- * Hill cipher

Transposition Techniques
(Change position)

- * Rail Fence cipher
- * Columnar Transposition cipher
- * Route cipher
- * Grille cipher

Similarities

- * Both are classical encryption methods a key for
- * Both use encryption & decryption
- * Both convert Plaintext to ciphertext
- * Both aim to provide Confidentiality.

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Differences

	Substitution Techniques	Transposition Techniques
Basic	Replace each letter with another letter or symbol	Rearrange the positions of letters without changing them
Effect on Data Structure	Changes the characters (content) of the data	Changes the order/position of characters
Example	A → D, B → E, C → F (Caesar Cipher)	HELLO → HLELO (Transposition)
Security Level	Generally less secure. Vulnerable to frequency analysis.	Generally more secure than simple substitution. Still breakable with known-plaintext attacks.

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Vulnerability Analysis

Substitution techniques are more vulnerable to attacks because they do not hide the pattern of letter positions, and frequency analysis can easily reveal the plaintext.

Transposition techniques only change the order, so if the attacker knows the method, it can be broken.

Steganography : Data Hiding , Watermarking & Survivability

Steganography (Hiding Secret Information)

1. Data Hiding

Conceals Secret information inside a cover medium so that no one except the intended receiver knows about it.

How it Contributes to Secure Communication

- * Keeps the existence of the message secret
- * Prevents unauthorized detection of message

Real-World Application

- * used by military to send secret messages hidden in images (or) videos.

2. Watermarking

Embeds information into digital media to prove ownership and protect copyright.

How it Contributes to Secure Communication

- * Ensures authenticity.
- * Protects from illegal copying.
- * helps in tracking ownership

Real-World Application

- * Copyright protection used in photos, videos to prove ownership of digital content

Steganography (Hiding Secret Information)

3. Survivability

Ability of hidden information to remain intact even after attacks or modifications on the cover medium.

How it contributes to Secure Communication

- * Ensures hidden data is not lost.
- * Provides robustness against compression, noise, cropping & other attacks

Real-World Application

- * Secure Image sharing
Hidden data remains even after the image editing or editing

Relationship

Data hiding provides Secrecy, Watermarking provides ownership & Survivability ensures durability of hidden information. Together they enable Secure Communication.

Together they ensure Confidentiality, Authenticity & Reliability of Information